MILITARY SPECIFICATION

NUT, SELF-LOCKING, A-286, 160 ksi Ftu, 450°F, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-locking A-286 corrosion resistant nuts for use in temperatures up to and including 450°F, in conjunction with expandable shank bolts conforming to MIL-B-85667.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2 1.1 Specifications and standards The following specifications and standards from a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-P-35	Passivation Treatments for Corrosion Resisting Steel
PPP-H-1581	Hardware (Fasteners and Related Items) Packaging and Packing for Shipment and Storage of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 53), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5310

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SPECIFICATIONS (continued)

MILITARY

MIL-H-6875	Heat Treatment of Steel (Aircraft Practice), Process for
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-W-8982	Wrenches, Splined, High Strength, Thin Wall, General Specification For
MIL-L-8937	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting, Nato Code Number, S-1738

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-410	Non-Destructive Testing Personnel Qualification and Certification
MIL-STD-1312	Fastener, Test Methods
MIL-STD-6866	Inspection, Liquid Penetrant
MS33588	Nut, Self-Locking, Aircraft, Design and Usage Requirements For
MS33787	Wrenching Element, External Spline, Dimensions For

(Copies of specifications, standards, handbooks, drawings and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46 1	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS 5525	Steel, Sheet, Strip & Plate, Corrosion & Heat Resistant
AMS 5737	Bars, Forgings & Tubing, Consumable Electrode Melted, 1650°F Solution & Precipitation Heat Treated

(Application for copies should be addressed to Society of Automotive Engineers, Inc. 400 Commonwealth Drive. Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein (except for associated detail specifications, specification sheets of MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. The nut furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids, see 4.3 and 6.3.

3.3 Retention of qualification. To maintain status on a Qualified Products List, certification shall be submitted at two year intervals to indicate continued compliance with the requirements of this specification, see 4.3.4.

3.4 Definitions. For definitions of terms used in this specification, see 6.4.

3.5 Materials.

3.5.1 Nut. The nut shall be fabricated from corrosion-resistant steel, AMS 5525 (A-286) as specified in the applicable specification sheet, and be capable of 160 KSI minimum ultimate tensile strength.

3.5.2 Washer. The washer shall be fabricated from corrosion-resistant steel (A-286) as specified in the applicable specification sheet.

3.6 Design. Nut design shall conform to the applicable specification sheet.

3.7 Construction. Nuts shall be of the prevailing torque-type. The locking device shall not operate by means of separate movement as a result of installation and shall not depend upon pressure on the bearing surface for locking action.

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3.7.1 Wrenching element. Wrenching element shall be in accordance with MS33787.

3.7.2 Bearing surface. After removal of the washer from the nut, the bearing surface shall be normal to the axis of the pitch diameter of the threads within the values specified on the applicable standard or drawing.

3.7.3 Threads. Unless otherwise specified, thread dimensions shall be in accordance with MIL-S-8879. Threads used on the locking device may be displaced in any manner which will provide self-locking nuts conforming to this specification. Thread gaging shall be performed after plating in accordance with the applicable thread specification, (except that the nut with locking device incorporated but without lubricant shall allow the go plug gage members to enter a minimum of three-fourths turn before engagement of the locking device). When lubricant prevents use of standard gages, the nut shall permit free rotational (finger torqued) bolt engagement of at least three-fourths turn.

3.8 Surface treatment. The nut shall be passivated in accordance with QQ-P-35.

3.9 Dry film lubricant. The nut shall be lubricated with dry film lubricant conforming to MIL-L-8937.

3.10 Axial strength. The nuts shall have a minimum axial strength of not less than the values specified in table I, see 4.5.2.

3.11 Torque.

3.11.1 Wrench torque. The nuts shall withstand the wrench torque values specified in table I, see 4.5.3.1. Upon completion of the test, the nut shall show no indications of permanent deformation which may prevent its proper application and removal with a socket-type wrench conforming to MIL-W-8982. The nuts shall show no evidence of cracking, see 4.5.7.

3.11.2 Torque effectivity, room temperature. The nuts shall not exceed the room temperature maximum locking torque nor produce less than the minimum breakaway torque specified in table III, see 4.5.3 2.

3.11.2.1 Torque effectivity, 450°F. The nuts shall not exceed the room temperature after bake maximum locking torque nor produce less than the minimum breakaway torque specified in table III, see 4.5.3.2.1.

3.11.3 Locking torque. The nuts shall not exceed the room temperature maximum locking torque specified in table III, see 4.5.3 3.

3.11.4 Permanent set. The nuts shall not exceed the maximum locking torque nor be less than the minimum breakaway torque specified in table III, see 4.5.3.4.

3.12 Accelerated vibration The nuts shall withstand the vibration test specified in 4.5.4.

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3.13 Stress durability. The nuts shall show no evidence of cracking, see 4.5.5.

TABLE I. Axial Strength and Wrench Torque.

Nominal Size	Cross Sectional Area (Sq-In)	Axial Strength Minimum (Lbs.)	Wrench Torque Minimum (In-Lbs.)
.1900-32	.0226	3,620	60
.2500-28	.0404	6,460	150
.3125-24	.0640	10,220	330
.3750-24	.0951	15,200	530
.4375-20	.1288	20,620	780
.5000-20	.1717	27,460	1,080
.5625-18	.2176	34,840	1,510
.6250-18	.2724	43,640	1,950
.7500-16	.3952	63,100	3,380
.8750-14	.5392	86,210	5,510
1.0000-12	.7027	111,990	7,550
1.1250-12	.9007	143,990	9,780
1.2500-12	1.1232	179,540	12,000
1.3750-12	1.3703	218,650	13,330
1.5000-12	1.6420	263,060	16,000

Axial Strength Values determined from Formula Wa = Ftu A;
Where A = Cross sectional area (Sq-In.) per NAS 1348.
Area based on maximum pitch diameter of bolt thread.
Ftu = 160 KSI Wa = Axial Strength (lbs.)

TABLE II. Stress Durability and Stress Corrosion.

Nominal Size	Nominal Grip Length of Bolt (In.)	Bolt Elongation (In.)
.1900-32	2.00	.0049
.2500-28	2.00	.0052
.3125-24	2.00	.0053
.3750-24	2.00	.0056
.4375-20	2.50	.0068
.5000-20	2.50	.0070
.5625-18	2.50	.0073
.6250-18	2.75	.0081
.7500-16	2.75	.0084
.8750-14	2.75	.0088
1.0000-12	3.00	.0096
1.1250-12	3.00	.0100
1.2500-12	3.00	.0103
1.3750-12	3.00	.0106
1.5000-12	3.00	.0109

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(a) Test bolt shall be stressed by torquing the nut until elongation is within 0.0005 of value specified. This elongation will indicate a stress of 50% of the minimum axial strength.

(b) Bolt elongations are calculated using bushing assembly lengths "A" (dimensions "L" per table VIII).

(c) Other bolt grip lengths permissible. Calculate elongation per MIL-STD-1312-14, using values for "A" equal to nominal grip length plus three times the thread pitch.

TABLE III. Torque Values.

Nominal Size	Seating Torque For Torque Effectivity Test (In-Lbs.)	Maximum Locking Torques 1/		Minimum Breakaway Torque (In-Lbs.)	Assembly Torque Vibration
		Room Temperature (In-Lbs.)	Room Temperature After Bake (In-Lbs.)		
.1900-32	56	18	36	2.0	32
.2500-28	130	30	60	3.5	53
.3125-24	290	60	120	6.5	107
.3750-24	480	80	160	9.5	142
.4375-20	700	100	200	14.0	177
.5000-20	980	150	300	18.0	267
.5625-18	1,360	200	400	24.0	356
.6250-18	1,780	300	600	32.0	533
.7500-16	3,020	400	800	50.0	
.8750-14	4,980	600	1,200	70.0	SEE
1.0000-12	6,840	800	1,600	90.0	4.5.4.1
1.1250-12	8,870	900	1,800	117.0	
1.2500-12	10,840	1,000	2,000	143.0	
1.3750-12	12,800	1,100	2,200	165.0	
1.5000-12	14,400	1,250	2,500	195.0	

1/ Values apply to nuts as lubricated by manufacturer.

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3.14 Stress corrosion. The nuts shall show no evidence of cracking when tested as specified in 4.5.6.

3.15 Discontinuities. Discontinuities in the nuts shall not exceed the depth limitations shown in table IV, see 4.5.7. Care shall be exercised not to confuse cracks with discontinuities, see 6.4.3. Cracks are not permitted in any location.

TABLE IV. Discontinuity Depth Limits, Maximum, Inch 1/.

Nominal Thread Size	Limits	Rejectable Criteria
.1900-32	.005	
.2500-28	.005	
.3125-24	.005	
.3750-24	.006	
.4375-20	.007	
.5000-20	.008	Any crack, regardless of location. Discontinuties in excess of specified limits.
.5625-18	.008	
.6250-18	.008	
.7500-16	.008	
.8750-14	.008	
1.0000-12	.008	
1.1250-12	.008	
1.2500-12	.008	
1.3750-12	.008	
1.5000-12	.008	

1/ Samples of nuts having indications may be sectioned and micro-examined to determine whether the indications are due to tool or die marks, or due to discontinuities, and to determine conformance of discontinuities to the depth limits specified in table IV.

3.16 Heat treatment. The nuts and washers shall be heat treated in accordance with MIL-H-6875 to develop the mechanical properties as specified herein.

3.17 Workmanship. Workmanship shall be consistent with the type or product, finish, and the class of thread fit specified. Sharp edges shall be broken; hanging burrs and slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections The inspection requirements specified herein are classified as follows:

a. Qualification Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 Qualification inspection

4.3.1 Sample. Qualification test samples (see table V) shall consist of 50 nuts for each size upon which qualification is desired. All bolts, screws, and mandrels necessary for tests specified herein shall be furnished by the manufacturer. Complete description and name of lubricating material used on the nuts shall be furnished. Samples shall be identified as required and forwarded to the activity responsible for qualification designated in the letter of authorization from that activity, see 6.3.

TABLE V. Qualification test plan.

Test	Paragraph	Nuts Required	Bolts or Studs Required
Examination of Product	4.5.1	All	0
Thread Fit	4.5.1	All	0
Bearing Surface Squareness	4 5 1.1	3	0
Axial Strength	4.5.2	5	5
Wrench Torque	4.5.3.1	5	5
Torque Effectivity, Room Temp. (Locking Element)	4.5.3.2	5	5
Torque Effectivity, 450°F (Locking Element)	4.5.3.2.1	5	5
Locking Torque	4.5.3.3	3	3
Permanent Set	4.5.3.4	3	3
Stress Corrosion	4.5.6	6	6
Accelerated Vibration	4.5.4	5	5
Stress Durability	4.5.5	6	6
Discontinuities	4.5.7	All	0

4.3.2 Inspections The qualification inspections shall consist of all the tests and examinations specified in 4.5.

4.3.3 Certified test report. The manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this specification. The test report shall include, as a minimum, actual results of the tests specified herein When this report is submitted, it shall be accompanied by a dated drawing which describes the manufacturer's product by specifying pertinent dimensions and tolerances, composition of material selected, coating and the heat treat. The manufacturer's part number for each size shall be included on the above drawing.

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4.3.4 Retention of qualification. Certification shall be requested by the Naval Air Development Center (NAVAIRDEVCEN), Code 6013, Warminster, PA 18974, from each manufacturer. NAVAIRDEVCEN will forward certification to the Naval Air Engineering Center (NAVAIRENGCEN), Code 5311, who is acting as agent for the Naval Air Systems Command (NAVAIR), AIR-53033G). Certification shall be at the time of the two year review and shall be signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified; (i.e., same process, materials, construction, design, manufacturer's part number, or designation); and meets the requirements of the current issue of the specification. Failure to provide the certification will be cause for removal from the QPL After completion of the certification review, the QPL will be reprinted to show the date of certification. (DD Form 1718, Certification of Qualified Products, shall be used for obtaining certification.)

4.4 Quality conformance inspection. The quality conformance inspections shall consist of those examinations and tests listed in 4.4.2 on samples selected in accordance with sampling plans A, B, and C.

4.4.1 Selection of samples. Sample nuts shall be selected at random from each inspection lot as specified herein.

4.4.1.1 Inspection lot A lot shall consist of finished nuts of the same type and diameter fabricated by the same process, from one heat of material, heat treated in the same manner, and produced as one continuous run, order, or part thereof.

4.4.1.2 Sampling Plan A. Sampling sizes for examination of product and thread fit (4.5.1) and bearing surface squareness (4.5.1.1) shall be in accordance with MIL-STD-105, inspection level S-1, as specified in table VI. The acceptance and rejection criteria shall be applied to the following acceptance quality levels (AQLs) pertaining to the corresponding class of characteristics:

Major AA	0.0 percent
Major A	0.4 percent
Major B	1.5 percent
Major C	2.5 percent
Major D	4.0 percent

4.4.1.2.1 Classification of defects. The classification of defects for self-locking nuts shall be:

Major AA

Visual inspection of package marking.

Major A

Visual presence of locking configuration.
Bearing surface squareness.

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Major B

Thread Fit.

Major C

Wrench Size - Height of wrenchable portion
Overall Height - Diameter of bearng surface.
Loose or hanging burrs - Concentricity of threads to base diameter.
All dimensional characteristics not covered above.

Major D

Part identification per applicable standard or drawing.

TABLE VI. Inspection Sampling Plan A

Lot Size	0.4% AQL			1.5% AQL			2.5% AQL			4.0% AQL		
	Sample Size	Ac	Re	Sample Size	Ac	Re	Sample Size	Ac	Re	Sample Size	Ac	Re
Under 26	25	0	1	8	0	1	5	0	1	3	0	1
26 thru 50	32	0	1	8	0	1	5	0	1	13	1	2
51 thru 90	32	0	1	8	0	1	20	1	2	13	1	2
91 thru 150	32	0	1	32	1	2	20	1	2	20	2	3
151 thru 280	32	0	1	32	1	2	32	2	3	32	3	4
281 thru 500	32	0	1	50	2	3	50	3	4	50	5	6
501 thru 1,200	125	1	2	80	3	4	80	5	6	80	7	8
1,201 thru 3,200	125	1	2	125	5	6	125	7	8	125	10	11
3,201 thru 10,000	200	2	3	100	7	8	200	10	11	200	14	15
10,001 thru 35,000	315	3	4	315	10	11	315	14	15	315	21	22
Over 35,000	500	5	6	500	14	15	500	21	22	315	21	22

Where sample size is equal to or greater than lot size 100% inspect.

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4.4.1.3 Sampling Plan B. For the axial strength test (4.5.2), locking torque test (4.5.3.3), and permanent set (4 5.3.4), the samples shall be selected in accordance with table VII.

TABLE VII. Sampling plan B.

Lot Size	Sample Size	Acceptance Number
Under 10,000	5	0
10,000 to 50,000	10	0
50,001 to 100,000	15	0
Over 100,000	27	1

4.4.1.4 Sampling Plan C. For stress durability (4.5.5), sample size shall be 5 for each lot and the acceptance number zero.

4.4.2 Tests. The quality conformance tests shall consist of:

a. Examination of Product (4.5.1 and 4.5.1.1)

b. Axial Strength (4.5.2)

c. Locking Torque (4.5.3.3)

d. Discontinuities (4.5.7)

4.5 Test methods.

4.5.1 Examination of product. The nuts shall be examined for conformance to this specification and applicable approved standards with respect to material, workmanship, dimensions, design and construction, and finish.

4.5.1.1 Bearing surface. After removal of washer from the nut bearing surface values shall be measured by means of a table squareness gage, with a seating surface diameter equal to or greater than the base diameter of the nut being examined, in conjunction with a feeler gage, with a class 3A threaded member at engagement in the locking device, and turned finger tight.

4.5.2 Axial strength. The nuts shall be assembled on test bolts as specified in table IX, having minimum grip length equal to three times the bolt diameter, and subjected to the axial strength load specified in table I. Test method and fixtures shall be in accordance with MIL-STD-1312-8. The nuts shall support the axial strength load without rupture. For qualification, apply load until nut or bolt failure occurs. For quality conformance, tests may be discontinued after minimum load has been achieved. Bolt failure below the rated axial strength of the nut due to defective bolt is not cause for rejection. In the event of such bolt failure, this test shall be repeated.

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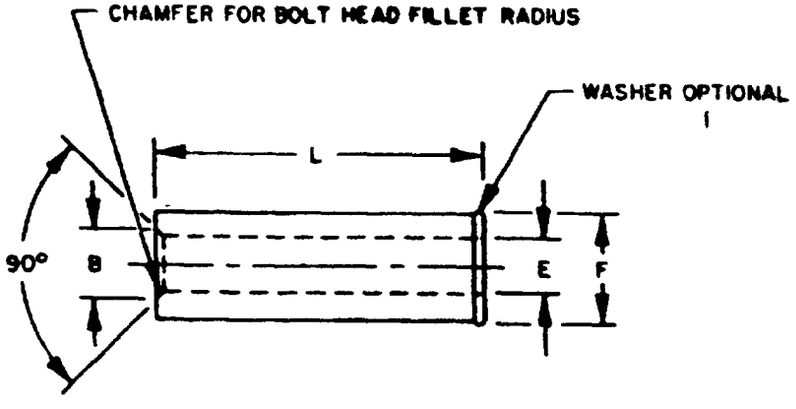
4.5.3 Torque.

4.5.3.1 Wrench torque. The nuts shall be installed on test bolts as specified in table IX with a steel fixture in accordance with figure 1, and torqued to the wrench torque test values specified in table I. The nuts shall then be removed and reinstalled to this torque fourteen additional times. Wrenches shall be of the socket type conforming to MIL-W-8982. Deformation which interferes with the proper application and removal of the nut with the wrench is sufficient cause for rejection.

4.5.3.2 Torque effectivity-room temperature. The nuts shall be installed on test bolts as specified in table IX with test fixtures in accordance with figure 1 and torqued to the seating torque values specified in table III. The nuts shall then be removed and reinstalled (seated) fourteen additional times. The maximum locking torque and minimum breakaway torque shall be that for the 1st, 7th and 15th cycles shall be recorded. The minimum breakaway torque shall be that torque required to start relative motion, between the nut and bolt, after the nut has been unseated and backed off one-half-turn.

4.5.3.2.1 Torque effectivity-after 450° bake. The nuts shall be installed on test bolts as specified in table IX with test fixtures in accordance with figure 1 and torqued to the seating torque values specified in table III. The assemblies shall then be baked for one hour at 450°F (-0 +25°) and cooled to room temperature. The nuts shall then be removed and reinstalled (seated and baked) four additional times. The maximum locking torque and minimum breakaway torques for the 1st and 5th cycles shall be recorded. The minimum breakaway torque shall be that torque required to start relative motion, between the nut and bolt, after the nut has been unseated and backed off one-half turn.

4.5.3.3 Locking torque. The nuts shall be installed and removed on test bolts, as specified in table IX, one time, recording maximum locking torque and minimum breakaway torque during the one cycle.



Test Fixture for Torque Effectivity and Wrench Torque.

Material: A286 CRES per AMS 5737

Heat Treat: Rockwell C32 Min

Ends of bushing and bushing plus washer to be flat and parallel within .002. Surface texture of ends: 32 microinches per ANSI B46.1.

FIGURE 1. Test Fixture.

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TABLE VIII. Dimensions for Torque Test Bushing Assembly.

Nut Nominal Thread Size (Ref.)	B Diameter ± .005	E Diameter +.000 -.010	F Diameter Minimum	L ±.010 (a)
.1900-32	.285	.205	Diameter of Nut Base or Twice the Diameter of the Bolt Whichever is Greater	2.125
.2500-28	.345	.265		2.125
.3125-24	.407	.327		2.125
.3750-24	.502	.390		2.125
.4375-20	.564	.452		2.625
.5000-20	.627	.515		2.625
.5625-18	.689	.577		2.625
.6250-18	.783	.640		2.875
.7500-16	.908	.765		2.938
.8750-14	1.033	.890		2.938
1.0000-12	1.158	1.015		3.188
1.1250-12	1.283	1.140		3.188
1.2500-12	1.440	1.265		3.188
1.3750-12	1.565	1.390		3.188
1.5000-12	1.690	1.565		3.188

(a) Length "L" may include one washer of equal material and hardness. Length "L" may vary to accommodate other bolt lengths - see table II, note c.

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TABLE IX. *Type of test bolt or stud.

Type	Test	Paragraph
I or III	Axial Strength, Room Temperature	4.5.2
I or III	Stress Durability	4.5.5
III	Torque Effectivity, Room Temperature (Locking Element)	4.5.3.2
I or III	Wrench Torque	4.5.3.1
III	Locking Torque	4.5.3.3
II	Permanent Set	4.5.3.4
III	Torque Effectivity, 450° (Locking Element)	4.5.3.2.1
III	Stress Corrosion	4.5.6
I or III	Accelerated Vibration	4.5.4

*Type I. Rolled threads in accordance with MIL-S-8879 thread form, having 160 KSI (minimum) strength level. If bolts are used, head configuration is optional

II. A. Maximum Stud - Cadmium plated 125 KSI (minimum) strength level with rolled threads in accordance with MIL-S-8879, except that the pitch diameter shall be established at 75%, +.0000-.0004 of the tolerance range of class 3A, above the minimum pitch diameter subject to check with functional pitch diameter tri-roll gages, see table X.

B. Minimum Mandrel - In accordance with figure 2. Threads shall conform to table X. It shall be checked with pitch diameter tri-roll gages.

Test bolts specified in this table are for referee purposes. Bolts or threaded studs may be used for routine testing. Threads may be rolled before or after heat treatment.

III Material shall be Titanium 6AL-4V rolled threads in accordance with MIL-S-8879 thread form having 160 KSI (minimum) strength level. If bolts are used, head configuration is optional.

4.5.3.4 Permanent set. Permanent set shall be subjecting the nut at room ambient temperature, to one complete installation and removal cycle on a maximum stud in accordance with table IX and then repeating the test cycle with the same clean nut on a minimum pitch mandrel, in accordance with table IX. Three of the nuts submitted shall be subjected to this test.

4.5.4 Vibration. Sample nuts with bolts of the size and quantities specified in table XI shall be vibrated in accordance with MIL-STD-1312-7. Vibration life, for each lot of five specimens tested in accordance with MIL-STD-1312-7, of less than 45,000 cycles shall be cause for rejection.

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TABLE X. Thread dimensions for maximum stud and minimum mandrel for permanent set test.

Maximum 3A Stud			Minimum 3A Mandrel			
Size	Pitch Diameter		+0.000 -0.0004 Major Diameter	+0.0000 -0.0004 Pitch Diameter	Tolerance in lead (per inch)	Tolerance on half angle of thread (in minutes)
	Maximum	Minimum				
.1900-32	.1691	.1687	.1840	.1674	±.0003	±15
.2500-28	.2261	.2257	.2435	.2243	±.0003	±15
.3125-24	.2847	.2843	.3053	.2827	±.0003	±15
.3750-24	.3471	.3467	.3678	.3450	±.0003	±15
.4375-20	.4042	.4038	.4294	.4019	±.0003	±15
.5000-20	.4667	.4663	.4919	.4643	±.0003	±15
.5625-18	.5255	.5251	.5538	.5230	±.0003	±10
.6250-18	.5880	.5876	.6163	.5854	±.0003	±10
.7500-16	.7084	.7080	.7406	.7056	±.0003	±10
.8750-14	.8275	.8671	.8647	.8245	±.0003	±10
1.0000-12	.9448	.9444	.9886	.9415	±.0003	±10
1.1250-12	1.0697	1.0693	1.1136	1.0664	± 0003	±10
1.2500-12	1.1947	1.1943	1.2386	1.1913	±.0003	±10
1.3750-12	1.3197	1.3193	1.3636	1.3162	±.0003	±10
1.5000-12	1.4447	1.4443	1.4886	1.4411	±.0003	±10

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TABLE XI. Vibration requirements.

Nut Size	Bolt Grip (Inch)	MIL-STD-1312-7 Bolts and Nuts Required (Min. of each) 450°F
.1900-32	1.125	10
.2500-28	1.188	10
.3125-24	1.188	10
.3750-24	1.188	10
.4375-20	1.188	10
.5000-20	1.188	10
.5625-18	1.188	10
.6250-18	1.188	10

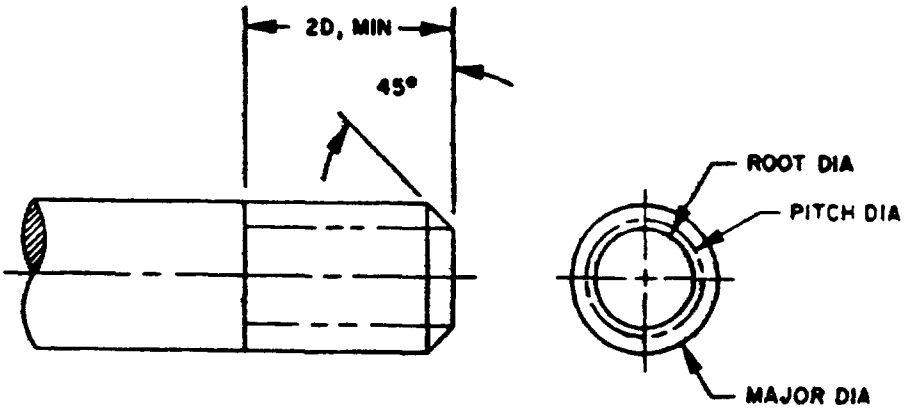
4 5.4.1 Vibration tests on nuts larger than 5/8-inch size are waived, provided that 5/8-inch nuts and smaller of the same type and design of locking device have satisfactorily passed the vibration test.

4.5.4.2 Preparation for vibration test to MIL-STD-1312-7.

4.5.4 3 Accelerated vibration. The nuts shall be assembled in accordance with figure 1 of MIL-STD-1312-7, with bolts and torque values as specified in table III. The nuts shall then be removed and reinstalled to this torque four additional times before being vibrated

4 5.4 4 Baking of test specimens Half of the nuts shall be assembled on the appropriate bolts and baked for six hours at 450°F. The baked specimens shall be allowed to cool slowly in air to room temperature. The nuts shall then be removed and reinstalled to torque values specified in table III for four additional times before being vibrated.

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DETAILS:

MATERIALS: STEEL

HARDNESS: HEAT TREATED TO ROCKWELL C60 TO 64

THREADS: THE THREAD FORM SHALL CONFORM TO THAT SHOWN IN MIL-S-8879.

THE THREADS SHALL BE RIGHT HAND. THE THREAD DIMENSIONS SHALL BE AS SHOWN IN TABLE X.

SURFACE ROUGHNESS: 20 MICROINCHES MAXIMUM IN ACCORDANCE WITH ANSI B46.1.

THE MANDREL END SHALL HAVE A 45-DEGREE CHAMFER EXTENDING BELOW THE ROOT DIAMETER. THE RESULTING SHARP FEATHER EDGE OF THE INCOMPLETE THREAD SHALL BE REMOVED BY STONING.

NOTE: CETYL ALCOHOL, TYPE I, GRADE A LUBRICANT PER MIL-L-87132 CAN BE USED ON MANDREL IF CORROSION RESISTANT STEEL NUTS ARE UNPLATED.

FIGURE 2. Mandrel.

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4.5.4.5 Method. Vibration shall be tested in accordance with MIL-STD-1312-7.

4.5.4.5.1 Determination shall be made throughout the test to guarantee that assembly is traversing the entire length of the slots in the test fixture. The test shall be run for 45,000 cycles except that it shall be stopped prior to the completion of the 45,000 cycles in the event a nut becomes disassembled from the bolt. The nut samples shall be examined under 10X magnification for cracks.

4.5.4.5.2 The nuts shall be considered to have failed to pass the vibration test under the following conditions:

a. If any structural failure, such as broken segments, locking inserts falling out, or cracks occur in the nuts during the test, provided failure is not the result of failure of the bolt.

b. If any nut comes completely off the bolt or can be turned completely on or off the bolt with the fingers during or after completion of 45,000 cycles.

c. If relative rotation between any nut and bolt exceeds 360 degrees.

4.5.5 Stress durability. Test in accordance with MIL-STD-1312-14, as specified in table VIII and torqued until the bolt elongates the amount specified in table II. The assemblies shall then be baked for 96 hours at 450°F (-0° + 25°) and cooled at room temperature. After removing nuts, polish bearing surface of nuts and examine at 10X to 50X magnification for cracks.

4 5.6 Stress corrosion. Test in accordance with MIL-STD-1312-9. The nuts shall be installed on test bolts as specified in table VIII and torqued until the bolt elongates the amount specified in table II. After 500 hours of alternate immersion, the parts shall be disassembled and the nut bearing surface shall be polished and examined, at 10X to 50X magnification, for cracks.

4.5.7 Cracks and discontinuities. Fluorescent penetrant inspection shall be used to determine the presence of cracks and discontinuities such as laps, seams and inclusions. Fluorescent penetrant indications of themselves shall not be cause for rejection. If indications are considered cause for rejection, representative samples shall be taken from these nuts showing indications and these samples shall be further examined by microexamination to determine whether the indicated discontinuities exceed the limits specified herein.

4.5.7.1 Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be performed in accordance with MIL-STD-6866. Personnel conducting fluorescent penetrant inspection shall be qualified in accordance with MIL-STD-410.

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5. PACKAGING

5.1 Self-locking nuts shall be preserved, packaged, packed, and marked for shipment in accordance with PPP-H-1581. Preservation and packaging shall be level A or C as specified in the contract or purchase order, see 6.2.

5.2 Packing shall be level A, B or C as specified in the contract or purchase order, see 6.2.

5.3 Marking of shipments. In addition to any special requirements of the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Military part number, see 3.1.

c. Applicable levels of packaging and packing.

6.2 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-85729) whether or not such products have actually been so listed by that date. Contractors' attention is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is (insert name and address of preparing activity) and information pertaining to qualification of products may be obtained from that activity.

6.2.1 Qualification tests will be authorized only upon presentation of certified test reports indicating that the nuts conform to this specification, see 4.3.3 and the applicable military specification sheet or a standard approved by the activity responsible for qualification.

6.3 Definitions applicable to the tests under this specification.

6.3.1 Installed. A nut is considered installed when a minimum of two thread pitches plus the chamfer of the male thread extend beyond the top of the nut.

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6.3.2 Removal cycle. The removal cycle is considered complete when the locking device is disengaged.

6.3.3 Crack. A crack is a clean break passing through the grain or grain boundary without inclusion of foreign elements.

6 3.4 Minimum breakaway torque. The minimum breakaway torque is that torque required to start nut rotation from a fixed position during a removal cycle with no load on the base of the nut.

6.3.5 Maximum locking torque. The maximum locking torque is the highest self-locking torque encountered in any installation or removal cycle with no load on the base of the nut.

6.4 Subject term (keyword) listing.

Nut
Self-Locking
Corrosion-resistant steel
160 KSI Ftu
450°F

Custodians:
Navy-AS
Air Force - 99
Army - AV

Reviewer Activity:
DLA - IS
Air Force - 82

Preparing activity:
Navy - AS
(Project No. 5310-1423)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-N-85729	NUT, SELF-LOCKING, A-286, 160 ksi Ftu, 450°F, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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